

Global, regional, and national prevalence of overweight and obesity in children and adults during 1980-2013: a systematic analysis for the Global B...

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BACKGROUND: In 2010, overweight and obesity were estimated to cause 3·4 million deaths, 3·9% of years of life lost, and 3·8% of disability-adjusted life-years (DALYs) worldwide. The rise in obesity has led to widespread calls for regular monitoring of changes in overweight and obesity prevalence in all populations. Comparable, up-to-date information about levels and trends is essential to quantify population health effects and to prompt decision makers to prioritise action. We estimate the global, regional, and national prevalence of overweight and obesity in children and adults during 1980-2013. **METHODS:** We systematically identified surveys, reports, and published studies (n=1769) that included data for height and weight, both through physical measurements and self-reports. We used mixed effects linear regression to correct for bias in self-reports. We obtained data for prevalence of obesity and overweight by age, sex, country, and year (n=19,244) with a spatiotemporal Gaussian process regression model to estimate prevalence with 95% uncertainty intervals (UIs). **FINDINGS:** Worldwide, the proportion of adults with a body-mass index (BMI) of 25 kg/m² or greater increased between 1980 and 2013 from 28·8% (95% UI 28·4-29·3) to 36·9% (36·3-37·4) in men, and from 29·8% (29·3-30·2) to 38·0% (37·5-38·5) in women. Prevalence has increased substantially in children and adolescents in developed countries; 23·8% (22·9-24·7) of boys and 22·6% (21·7-23·6) of girls were overweight or obese in 2013. The prevalence of overweight and obesity has also increased in children and adolescents in developing countries, from 8·1% (7·7-8·6) to 12·9% (12·3-13·5) in 2013 for boys and from 8·4% (8·1-8·8) to 13·4% (13·0-13·9) in girls. In adults, estimated prevalence of obesity exceeded 50% in men in Tonga and in women in Kuwait, Kiribati, Federated States of Micronesia, Libya, Qatar, Tonga, and Samoa. Since 2006, the increase in adult obesity in developed countries has slowed down. **INTERPRETATION:** Because of the established health risks and substantial increases in prevalence, obesity has become a major global health challenge. Not only is obesity increasing, but no national success stories have been reported in the past 33 years. Urgent global action and leadership is needed to help countries to more effectively intervene. **FUNDING:** Bill & Melinda Gates Foundation. Copyright © 2014 Elsevier Ltd. All rights reserved.